Bayesian Network

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Summary: Deterministic Search

- Search for Planning
  - Goal: Find a sequence of actions to reach a goal state
  - Assumption: Environment is fully known and deterministic
  - Examples: BFS, DFS, A* search
- Search for Assignment (Constraint Satisfaction Problems)
  - Goal: Assign values to variables under constraints
  - Assumption: Constraints are fixed and deterministic
  - Examples: Backtracking, Forward Checking, Arc Consistency

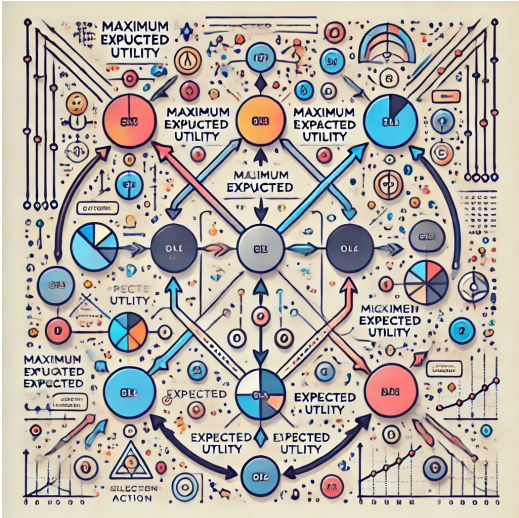
Summary: Deterministic Search

- Adversarial Search
  - Goal: Choose optimal moves in competitive settings
  - Assumption: Opponent is rational and environment is deterministic
  - Examples: Minimax, Alpha-Beta Pruning

- All three types of search assume a fully observable and deterministic world. But real-world problems often involve uncertainty, randomness, and incomplete information.
  - This is where Bayesian Networks come in: modeling and reasoning under uncertainty.

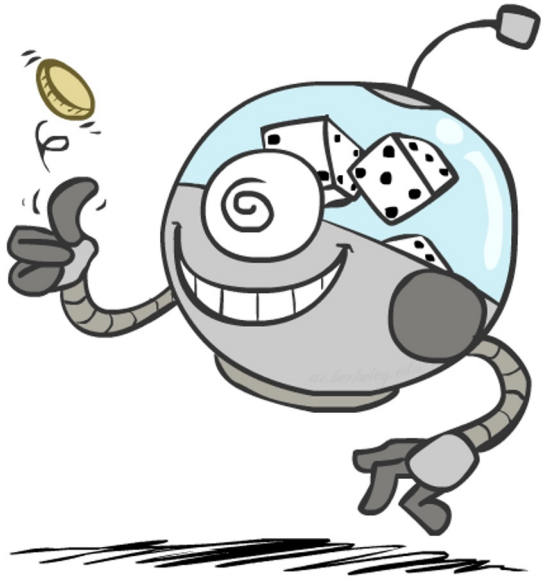
Maximum Expected Utility (MEU)

- An agent is considered rational if and only if it selects the action that maximizes its expected utility, calculated as the weighted average over all possible outcomes of that action.



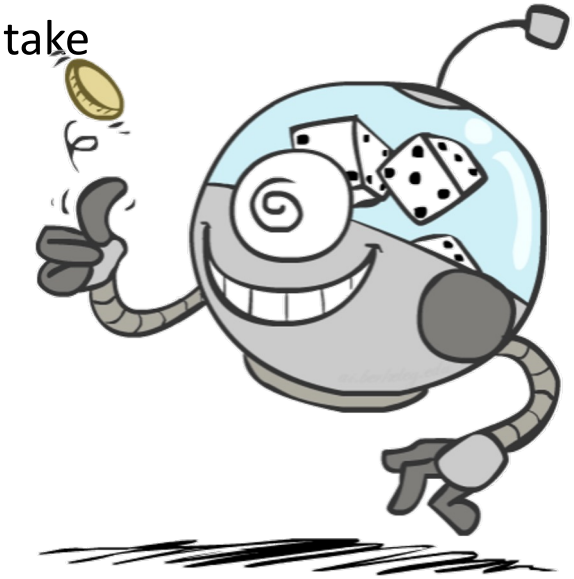
Random Variable

- Random variable is some aspect of the world about which we (may) have uncertainty.
  - R = Is it raining?
  - T = Is it hot?
  - D = How long will it take to drive to work?



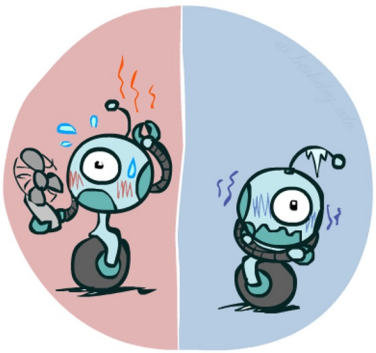
Random Variable

- Notations:
  - Capital letters: Random variables
  - Lowercase letters: values that the random variable can take
    - $r \in \{+r, -r\}, or\ \{rain, sun\}$
    - $t \in \{+t, -t\}, or\ \{hot, cold\}$
    - $d \in [0, \infty)$



Probability Distribution

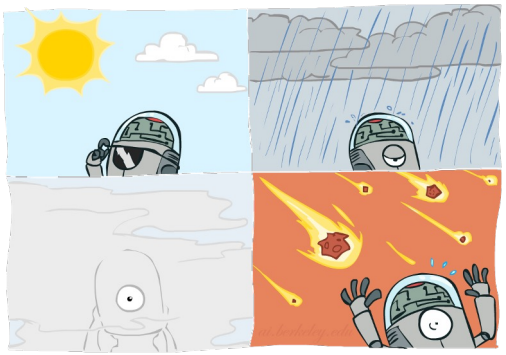
- We can associate a probability with each value.



$P(T)$

T	P
hot	0.5
cold	0.5

Temperature (T)



$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Weather (W)

Joint Distribution

- A joint distribution over a set of random variables $X_1, X_2, \ldots X_n$ specifies a real number for each assignment:

$$P(X_1 = x_1, X_2 = x_2, \ldots X_n = x_n)$$

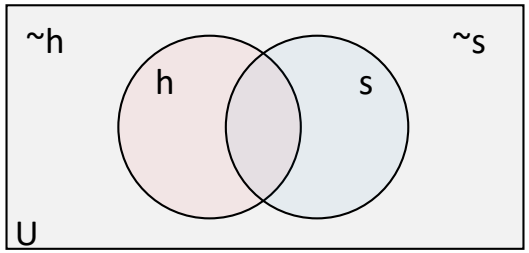
or $$P(x_1, x_2, \ldots x_n)$$

- It must obey:
  - Non-negativity $P(x_1, x_2, \ldots x_n) \geq 0$
  - Normalization $\sum_{(x_1, x_2, \ldots x_n)} P(x_1, x_2, \ldots x_n) = 1$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Visualization: Joint Distribution



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

What is the size of distribution if considering n variables with domain sizes d?

- For all but the smallest distributions, impractical to write out!

Marginal Distribution

- Marginal distributions are sub-tables which eliminate variables.
  - Marginalization (summing out): Combine collapsed rows by adding.

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Summing out W →

$P(T)$

T	P
hot	0.5
cold	0.5

Summing out T →

$P(W)$

W	P
sun	0.6
rain	0.4

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Conditional Distribution

- Conditional distributions are probability distributions over some variables given fixed values of others.

Joint Distribution

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Conditional Distributions

$$P(W|T)$$

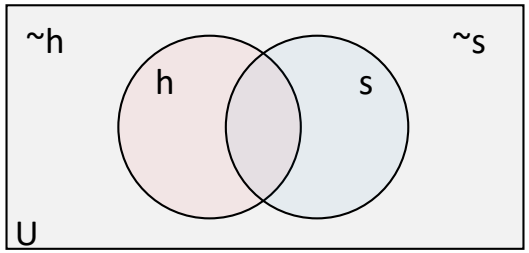
$$P(W|T = hot)$$

W	P
sun	0.8
rain	0.2

$$P(W|T = cold)$$

W	P
sun	0.4
rain	0.6

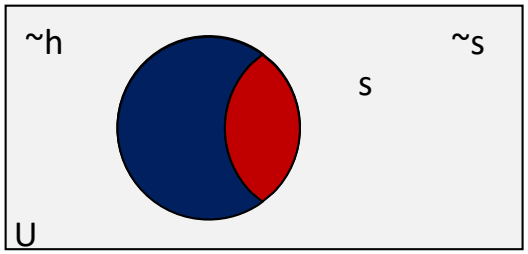
Visualization: Marginal/Conditional Distributions



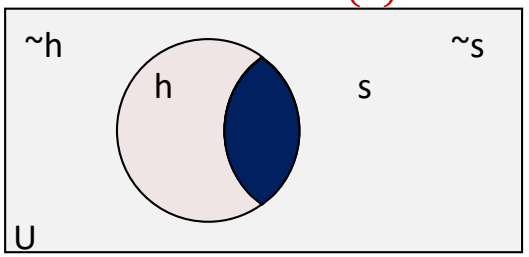
$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(h) = P(h, s) + P(h, \sim s)$$



$$P(s|h) = \frac{P(s, h)}{P(h)}$$



Normalization Trick

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)}$$
$$= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)}$$
$$= \frac{0.2}{0.2 + 0.3} = 0.4$$

$$P(W = r|T = c) = \frac{P(W = r, T = c)}{P(T = c)}$$
$$= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)}$$
$$= \frac{0.3}{0.2 + 0.3} = 0.6$$

$$P(W|T = c)$$

W	P
sun	0.4
rain	0.6

Normalization Trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence

$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)

$P(W|T = c)$

W	P
sun	0.4
rain	0.6

Probabilistic Inference

- Probabilistic inference: compute a desired probability based on other known probabilities.
- Probabilities can change when new evidence is observed:
  - P(on time | no accidents, 5 a.m.) = 0.95;
  - P(on time | no accidents, 5 a.m., raining) = 0.80;
  - Observing new evidence (e.g., raining) updates our belief about the likelihood of being on time.

Inference by Enumeration

- P(W)?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Full joint distributions

Inference by Enumeration

- P(W)?

  P(sun)=.3+.1+.1+.15=.65

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Full joint distributions

Inference by Enumeration

- P(W)?

  P(sun)=.3+.1+.1+.15=.65
  P(rain)=1-.65=.35

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Full joint distributions

Inference by Enumeration

- P(W)?

  P(sun)=.3+.1+.1+.15=.65
  P(rain)=1-.65=.35

- P(W | winter, hot)?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Full joint distributions

Inference by Enumeration

- P(W)?

  P(sun)=.3+.1+.1+.15=.65
  P(rain)=1-.65=.35

- P(W | winter, hot)?

  P(sun|winter,hot)~.1
  P(rain|winter,hot)~.05

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Full joint distributions

Inference by Enumeration

- P(W)?

  P(sun)=.3+.1+.1+.15=.65
  P(rain)=1-.65=.35

- P(W | winter, hot)?

  P(sun|winter,hot)~.1
  P(rain|winter,hot)~.05
  P(sun|winter,hot)=2/3
  P(rain|winter,hot)=1/3

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Full joint distributions

Inference by Enumeration

General case:

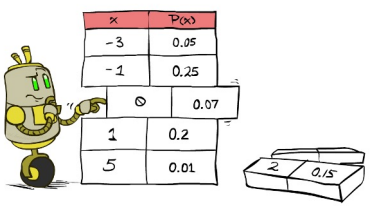
- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
- Query variable: Q
- Hidden variables: $H_1 \dots H_r$

$$\left.\begin{array}{l} E_1 \dots E_k = e_1 \dots e_k \\ Q \\ H_1 \dots H_r \end{array}\right\} \begin{array}{l} X_1, X_2, \dots X_n \\ \textit{All variables} \end{array}$$

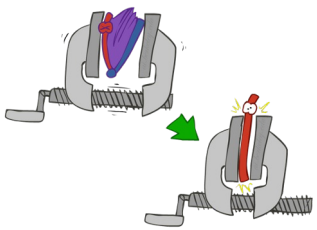
We want:

$$P(Q|e_1 \dots e_k)$$

Step 1: Select the entries consistent with the evidence



Step 2: Sum out H to get joint of query and evidence



Step 3: Normalize

$$\times \frac{1}{Z}$$

$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots X_n}$$

$$Z = \sum_q P(Q, e_1 \cdots e_k)$$

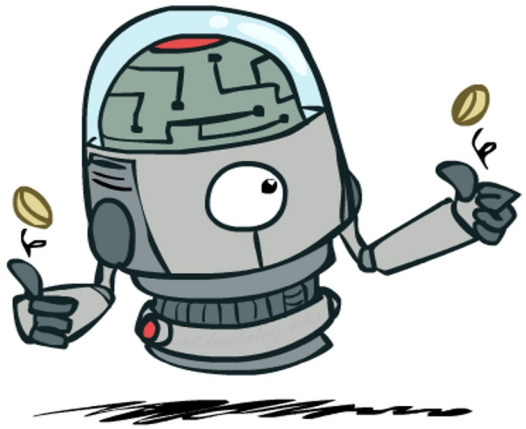
$$P(Q|e_1 \cdots e_k) = \frac{1}{Z} P(Q, e_1 \cdots e_k)$$

Independence

- Two variables are independent if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

  - This says that their joint distribution factors into a product of two simpler distributions.
  - Another form: $\forall x, y : P(x|y) = P(x)$
  - We write: $X \perp\!\!\!\perp Y$



Independence is a simplifying modeling assumption.

Example: Independence?

$P(T)$

T	P
hot	0.5
cold	0.5

$P_1(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P_2(T, W)$

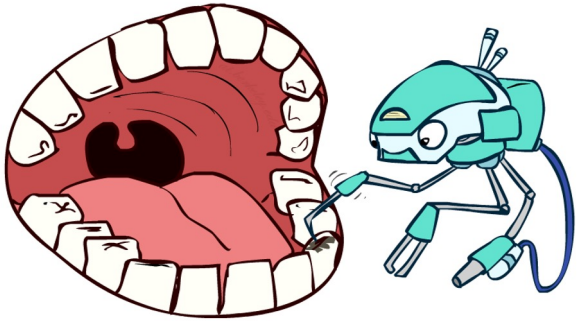
T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

$P(W)$

W	P
sun	0.6
rain	0.4

Conditional Independence

- P(Toothache, Cavity, Catch)?
  - If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
    - P(+catch | +toothache, +cavity) = P(+catch | +cavity)
  - The same independence holds if I don’t have a cavity:
    - P(+catch | +toothache, -cavity) = P(+catch| -cavity)
  - Therefore, Catch is **conditionally independent** of Toothache given Cavity:
    - P(Catch | Toothache, Cavity) = P(Catch | Cavity)
    - OR P(Toothache | Catch , Cavity) = P(Toothache | Cavity)
    - OR P(Toothache, Catch | Cavity) = P(Toothache | Cavity) P(Catch | Cavity)



Conditional Independence

- Unconditional (absolute) independence is very rare.
- Conditional independence is our most basic and robust form of knowledge about uncertain environments.
  - *X* is conditionally independent of *Y* given *Z* $X \perp\!\!\!\perp Y | Z$

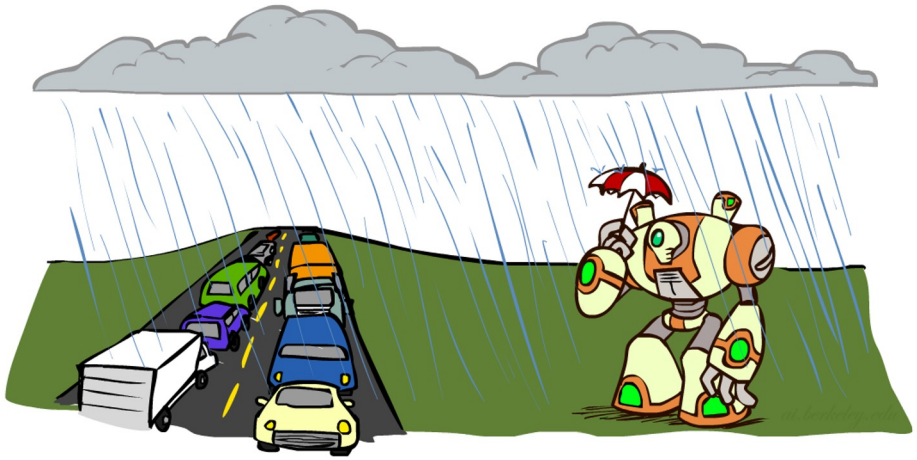
    if and only if: $\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$

    or, equivalently, if and only if $\forall x, y, z : P(x|z, y) = P(x|z)$

$$P(x|z, y) = \frac{P(x, z, y)}{P(z, y)} = \frac{P(x, y|z)P(z)}{P(y|z)P(z)} = \frac{P(x|z)P(y|z)P(z)}{P(y|z)P(z)}$$

Conditional Independence

- What about this domain:
  - Traffic
  - Umbrella
  - Raining



Conditional Independence and Chain Rule

- Chain rule:

$$P(X_1, X_2, \ldots X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)\ldots$$

- Trivial decomposition:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

- With assumption of conditional independence:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

Bayes Rule

- The Bayes rule underlies most modern AI systems for probabilistic inference:

$$\mathbf{P}(Y \mid X) = \frac{\mathbf{P}(X \mid Y)\mathbf{P}(Y)}{\mathbf{P}(X)}$$

- A more general version conditionalized on some background evidence or hidden variable $\mathbf{e}$:

$$\mathbf{P}(Y \mid X, \mathbf{e}) = \frac{\mathbf{P}(X \mid Y, \mathbf{e})\mathbf{P}(Y \mid \mathbf{e})}{\mathbf{P}(X \mid \mathbf{e})}$$

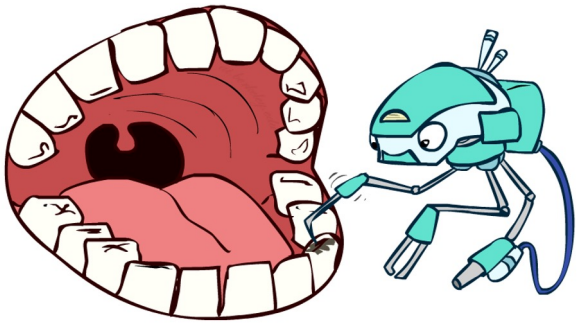
Bayes Rule

- We can make inference by the Bayes Rule

$$P(cause \mid effect) = \frac{P(effect \mid cause)P(cause)}{P(effect)}$$

  - Example: cause can be a disease, effect can be a symptom.
  - The inferring by the Bayes rule is then similar to the process of making a diagnosis.

P(Cavity | Toothache , Catch)?



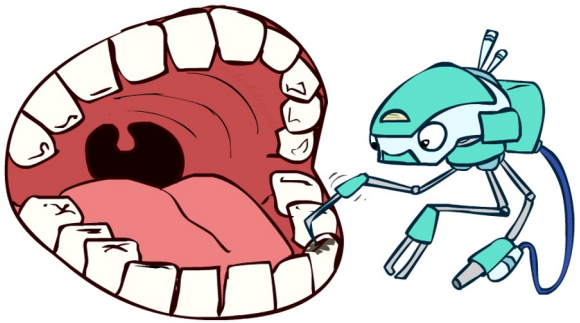
Bayes Rule

	toothache		*¬toothache*	
	catch	*¬catch*	*catch*	*¬catch*
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

Full joint distrubtion: usually difficult to model

P(Cavity | Toothache , Catch) ∝ P(Toothache, Catch | Cavity) * P(Cavity)

= P(Toothache | Cavity) * P(Catch | Cavity)
* P(Cavity)



Likelihood and Posterior

$$\mathbf{P}(Y \mid X) = \frac{\mathbf{P}(X \mid Y)\mathbf{P}(Y)}{\mathbf{P}(X)}$$

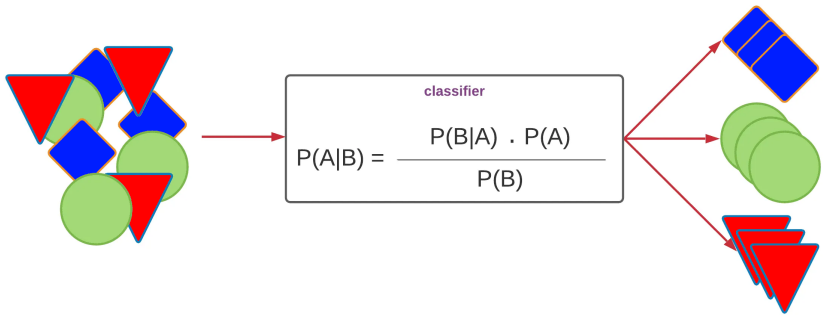
- $\mathbf{P}(Y|X)$: posterior probability of X given Y.
- $\mathbf{P}(X|Y)$: likelihood of Y given X.
- $\mathbf{P}(X)$: prior probability of X.
- $\mathbf{P}(Y)$: prior probability of Y.

X: can be effect (i.e., data) that is observable.
Y: can be a cause (i.e., hypothesis) underlying the effect.

- **Posterior** probability is the probability of the hypothesis being true given the data. It updates our belief about the hypothesis after collecting new data. The posterior probability is what we're ultimately interested in calculating.
- **Likelihood** is the probability of the data given that the hypothesis is true. It quantifies how likely we are to observe the data under the assumption that the hypothesis holds.

Naïve Bayes Model

- When a single cause directly influences a number of effects, all of which are conditionally independent, the full joint distribution is called a naïve Bayes model.

$$\mathbf{P}(Cause, Effect_1, \ldots, Effect_n) = \mathbf{P}(Cause) \prod_i \mathbf{P}(Effect_i \mid Cause)$$

- It is naturally a naïve Bayes classifier, if the cause can be discretized.